

# Digital Calibration Note for OPH Screen Microphysics: The Octahedral $\mathbb{Z}_2/S_3$ Patch Simulator

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May 17, 2026

**Paper release: r1454   Released: May 17, 2026**

## Abstract

This note defines the octahedral  $\mathbb{Z}_2/S_3$  simulator ladder as a digital calibration surface. The simulator gives an exact finite patch cover, explicit overlap collars, controlled defects, record writes, repair schedules, edge-sector labels, and negative controls. Its scope is fixed-cutoff interface logic and sphere-geometry charts. It is not the physical microphysics picture of OPH and should not be read as evidence that the universe is a literal spherical quantum computer.

## 1 Reading Rule

The microphysics paper *Federated Echosaedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH* treats the spherical screen as an observer-facing regulator chart and the implementation surface as a federation of finite observer patches.

This note has a narrower role:

1. validate finite patch, overlap, record, and repair bookkeeping;
2. test frustrated-cycle and defect behavior in an exact digital setting;
3. calibrate the edge-sector law on a finite declared interface;
4. provide reproducible benchmark names and run-manifest expectations;
5. avoid any claim that the digital sphere is the literal OPH substrate.

## 2 Octahedral Patch Cover

Use the octahedral cellulation of  $S^2$  as a small digital regulator chart. Label vertices

$$X_+, X_-, Y_+, Y_-, Z_+, Z_-.$$

The graph has twelve edges and eight triangular faces. The patch cover is the three-patch cover

$$P_X, \quad P_Y, \quad P_Z,$$

where  $P_X$  contains the faces incident on  $X_+$  or  $X_-$ , and similarly for  $P_Y$  and  $P_Z$ . The pairwise overlaps are denoted

$$B_{XY}, \quad B_{XZ}, \quad B_{YZ}.$$

The cover is intentionally small. It is large enough to expose pairwise overlaps, repair schedules, and a frustrated cycle, while remaining small enough for exact state and log inspection.

### 3 $\mathbb{Z}_2$ Pilot

The first digital pilot uses  $G = \mathbb{Z}_2$ . Each oriented link carries one bit or one qubit in the computational basis. The local gauge constraint is the even-parity constraint on incident links. Face holonomies are mod-two sums around triangular faces.

The pilot contains:

1. link registers on the twelve octahedral edges;
2. face-holonomy readouts;
3. one coarse cut bit for each active overlap;
4. record registers for patch-local write/compare/verify events;
5. a finite repair menu that flips declared collar links when doing so lowers the mismatch score without violating the local acceptance rule.

The  $\mathbb{Z}_2$  pilot is the abelian sanity surface. It should pass clean synchronization cases, expose stale records, repair single collar flips, and refuse to hide frustrated cycles.

### 4 $S_3$ Upgrade

The first nonabelian upgrade keeps the octahedral graph and patch cover. The cut bit becomes an encoded  $S_3$  register. Each group element is encoded in three bits with an invalid-word penalty. The vertex projector is

$$P_v^{S_3} = \frac{1}{6} \sum_{h \in S_3} U_v(h),$$

where  $U_v(h)$  is the local left/right action on links incident to  $v$ .

The upgrade tests:

1. nonabelian orientation bookkeeping;
2. conjugacy-class sector labels;
3. invalid-word penalties;
4. noncommuting defect order;
5. class-sensitive edge-sector readout.

Passing the  $S_3$  suite does not prove compact-gauge reconstruction. It only shows that the fixed-cutoff interface can carry nonabelian labels and record their repair behavior without collapsing to the abelian pilot.

## 5 Benchmark Families

The named calibration gates are:

1. **Z2-0-clean**: initialize the  $\mathbb{Z}_2$  pilot in the clean sector and verify stable synchronization.
2. **Z2-1-stale-record**: flip one stored record after a write layer and require the next compare/repair cycle to expose it.
3. **Z2-2-collar-flip**: inject one collar-edge flip before compare and require local repair.
4. **Z2-3**: double-overlap test. Inject two collar defects on different overlaps and test schedule-insensitive repair.
5. **Z2-4**: frustrated-cycle test. Inject a cyclic obstruction and require the simulator not to fake global agreement.
6. **S3-0-clean**: initialize the  $S_3$  build in the declared identity sector.
7. **S3-1-transposition**: inject a transposition-class defect.
8. **S3-2-threecycle**: inject a three-cycle defect.
9. **S3-3-mixed-class-cycle**: inject mixed nonabelian class data around the cycle and require class-sensitive failure reporting.

## 6 Run Manifest

Each digital run should write a manifest containing:

1. benchmark name and simulator commit;
2. graph, patch cover, group, encoding, and invalid-word policy;
3. repair weights, tie-break rule, and schedule;
4. per-cycle link state, face holonomies, overlap labels, record values, mismatch score, and repair move;
5. final normal form or obstruction report;
6. pass/fail decision with declared failure class.

The manifest is a digital analogue of the hardware evidence-bundle rule. It makes calibration claims auditable without confusing them with hardware evidence.

## 7 Negative Controls

Mandatory negative controls include:

1. stale-record control: disable record refresh and require the stale-record benchmark to fail;
2. wrong-orientation control: reverse one  $S_3$  edge orientation and require a nonabelian failure signature;

3. no-illegal-penalty control: set the invalid-word penalty to zero and require failure in the  $S_3$  suite;
4. fake-repair control: allow repair moves that rewrite records without touching overlap data and require the exact compare log to reject the run.

## 8 Boundary

This calibration note supports the fixed-cutoff theorem interface by making it executable in a small finite regulator chart. It does not identify the unique OPH UV completion, does not carry hardware evidence, does not prove the Yang–Mills repair-gap theorem, and does not promote source-only hadron predictions.